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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2024

EC2272-1- DATA STRUCTURES AND ALGORITHMS

(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following header files must necessarily be included to use dynamic memory allocation functions?
A) `stdlib.h` B) `memory.h`
C) `dos.h` D) `stdio.h`
2. The `free()` function frees the memory state pointed to by a pointer and returns _____.
A) the same pointer B) the memory address
C) no value D) an integer value
3. Which data structure is used for evaluating postfix expression?
A) Queue B) Stack
C) B-Trees D) Linked-list
4. Evaluate the postfix expression $ab + cd/-$ where $a=5, b=4, c=9, d=3$.
A) 23 B) 15
C) 6 D) 10
5. A linear list of elements in which deletion can be done from one end and insertion can take place only at the other end is known as _____.
A) Queue B) Stack
C) Tree D) Linked list
6. Which of the following information is stored in a doubly-linked list's nodes?
A) Value of node B) Address of next node
C) Address of prev node D) All the three information
7. What would be the Prefix notation for the given equation?
 $A+(B*C)$
A) $i) +A*CB$ B) $*B+AC$
C) $+A*BC$ D) $*A+CB$
8. Consider the following operation performed on a stack of size 5.
`Push(1); Pop(); Push(2); Push(3); Pop(); Push(4); Pop(); Pop(); Push(5);`
After the completion of all operation, the number of elements present in stack is?
A) 1 B) 2
C) 3 D) 4
9. Which of the following is a self-balancing binary search tree?
A) AVL Tree B) Binary Tree
C) General Tree D) Complete Binary Tree

10. What is a complete binary tree?
- A) A binary tree with no leaf nodes
B) A binary tree in which all nodes have exactly two children
C) A binary tree where nodes are stored in a linked list
D) A binary tree in which all levels are completely filled except possibly the last level
11. What is the main advantage of threaded binary trees?
- A) To make tree traversal faster
B) To reduce memory usage
C) To improve search efficiency
D) To remove null pointers
12. In graph theory, what is the minimum number of edges required to connect all vertices in a spanning tree?
- A) $n+1$
B) $n-1$
C) $n-2$
D) n
13. In which type of tree every node has either 0 or 2 children?
- A) Complete Binary Tree
B) AVL Tree
C) Full Binary Tree
D) B Tree
14. What is the degree of a vertex in a graph?
- A) The number of edges incident on the vertex
B) The number of connected vertices
C) The total weight of the edges connected to the vertex
D) The number of leaf nodes
15. In which type of graph does every pair of vertices have a path between them?
- A) Disconnected Graph
B) Complete Graph
C) Directed Graph
D) Undirected Graph
16. What is the time complexity of the binary search algorithm in the average case?
- A) $O(n)$
B) $O(\log n)$
C) $O(n \log n)$
D) $O(1)$
17. In quick sort, what is the worst-case time complexity?
- A) $O(n)$
B) $O(n \log n)$
C) $O(n^2)$
D) $O(\log n)$
18. Which of the following problems is solved using a greedy algorithm?
- A) Single source shortest path
B) Bubble Sort
C) Binary Search
D) Merge Sort
19. In dynamic programming, how are subproblems solved?
- A) Recursively
B) Iteratively
C) By storing results of solved subproblems
D) By brute force
20. Which of the following is NOT an application of dynamic programming?
- A) Chain Matrix Multiplication
B) Traveling Salesman Problem
C) Dijkstra's Algorithm
D) Longest Common Subsequence

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain how arrays are represented in memory. Write a C program to access array using pointers.	8	L*2	1	1,2
b)	Develop a C program to delete an element from an array at given position. Explain the concept/logic with an example.	8	L3	1	1,2
2. a)	Evaluate the following postfix expressions (stepwise) considering each digit as an operand. i. $2\ 3\ 4\ * +$ ii. $3\ 4\ * 2\ 5\ * +$ iii. $2\ 3\ 1\ * + 9 -$	8	L3	2	1,2
b)	Implement enqueue, dequeue and display operations on linear queue using arrays in C programming Language.	8	L3	2	1,2

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| 3. a) | Show how to initialize strings and perform string operations in C language. | 8 | L2 | 1 | 1,2 |
| b) | Develop a C program to perform insertion and deletion of a node in a singly linked list. | 8 | L3 | 2 | 1,2 |

Unit – II

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|-------|---|---|----|---|-----|
| 4. a) | Explain in order, and preorder and post order traversals in a tree. Write recursive C functions for in order, and preorder and post order traversals of a binary tree. | 8 | L3 | 3 | 1,2 |
| b) | Implement the adjacency matrix representation of an undirected graph using C language and explain with an example. | 8 | L3 | 3 | 1,2 |
| 5. a) | Provide an example for undirected, connected graph with at least 4 vertices and write all the possible spanning trees. Also show how to find the minimum spanning tree. | 8 | L3 | 3 | 1,2 |
| b) | Demonstrate how to traverse a graph using Depth First Search (DFS) algorithm and explain with suitable example. | 8 | L3 | 3 | 1,2 |
| 6. a) | Describe the steps in developing an algorithm with an example problem | 6 | L2 | 4 | 1,2 |
| b) | Given two algorithms for a task, how do you find out which one is better? Explain with an example. | 4 | L2 | 4 | 1,2 |
| c) | Analyze the time complexity of the "for loop" for different scenarios. | 6 | L2 | 4 | 1,2 |

Unit – III

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|-------|---|---|----|---|-----|
| 7. a) | Apply merge sort to sort the following elements and illustrate its time complexity.
50 ,5 ,100 ,6 ,8 ,1 ,23 ,13 ,6 ,11 | 8 | L3 | 5 | 1,2 |
| b) | Write an algorithm/program to find the single source shortest path using greedy approach (Dijkstra's Algorithm). | 8 | L3 | 5 | 1,2 |
| 8. a) | Design an algorithm/program to perform quick sort on an array with an example. | 8 | L3 | 5 | 1,2 |
| b) | Discuss the role of dynamic programming in solving i) longest common subsequence and ii) chain matrix multiplication, and explain how it is different from greedy solutions | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
